

Parent Guide — senior / module-14- magic-squares

IKS Curriculum

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Parent Guide — Magic Squares — Bhadragaṇita (Senior)

What is your child learning?

A magic square is an $n \times n$ grid of distinct integers where every row, column, and main diagonal sums to the same constant. Nārāyaṇa Paṇḍita's *Gaṇita-kaumudī* (1356 CE), Book 14 (*Bhadra-gaṇita*), gives the most systematic pre-modern Indian treatment, including the stair-step (kothumi) method for odd-order squares. Cross-cultural comparison with Lo Shu (China) and Dürer (Europe) shows the universality of the puzzle and the Indian contribution to its mathematical systematisation.

Why this matters

This module is part of an Indian Knowledge Systems curriculum that introduces classical Indian frameworks alongside their modern-scientific parallels. Your child will learn to (a) name and use precise Sanskrit terms, (b) understand the observational basis for the classical framework, and (c) relate it to current scientific or scholarly understanding.

What you can do at home

Your child is learning to construct magic squares — grids where every row, column, and diagonal sums to the same number. It's a puzzle and a piece of cultural mathematics. Ask them to show you the 3×3 with 1–9 — it's the foundational case.

Conversation prompts

Try one of these over a meal or walk:

1. “What’s one Sanskrit word you learned this week? What does it mean?”
2. “What did you find surprising in this module?”
3. “Is there anything you’re not sure about? Let’s look it up together.”
4. “How would you explain this to your grandmother / grandfather?”

Questions parents often ask

Q: Is this religious instruction? A: No. The module presents Indian classical frameworks (cosmology, mathematics, astronomy, ecology) as historical knowledge systems with their own logic and method. The framing is scholarly and comparative, not devotional.

Q: Will my child be tested on Sanskrit? A: Sanskrit terms are introduced for precision, not for rote memorisation. The summative quiz asks students to *use* the terms — for example, “Which doṣa is dominant in the morning hours?” — rather than translate them.

Q: How does this connect to NCERT / board syllabus? A: This module supplements (does not replace) NCERT content. NEP 2020 §4.6 calls for the integration of Indian Knowledge Systems. The pack’s README.md lists the specific NEP principles each module advances.

Reading further (for parents)

See sources.md in this pack for the full source list.

10-Day Lesson Plan

Lesson Plan — Magic Squares — Bhadragaṇita (Senior)

10-day arc, 45 min per session.

Day 1 — Discover the 3×3

Objective: Build the unique 3×3 magic square with 1–9 by trial.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 2 — The Magic Constant

Objective: Derive why 3×3 with 1–9 must sum to 15 per row.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 3 — How Many 3×3 Magic Squares?

Objective: Count the equivalence classes (up to symmetry).

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 4 — Stair-Step Method (Nārāyaṇa)

Objective: Apply Nārāyaṇa’s algorithm to build a 5×5 square.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 5 — Practice — 5×5 and 7×7

Objective: Use the method on multiple odd-order squares.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 6 — 4×4 and the Khajuraho Square

Objective: Examine the Khajuraho inscription and its panel-magic property.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 7 — Historical Context — Nārāyaṇa and Predecessors

Objective: Place *Gaṇita-kaumudī* in 14th c. Indian math.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 8 — Cross-Cultural — Lo Shu and Dürer

Objective: Compare Indian, Chinese (Lo Shu), and European (Dürer 1514) magic squares.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 9 — Design Your Own

Objective: Create a personalised 4×4 or 5×5 magic square as visual art.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 10 — Final Share and Assessment

Objective: Present design + summative quiz.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Pacing notes

- Senior sessions are 45 min. Adjust if your block is different.
- Days 5 and 9 are buffer days — use to reinforce weaker concepts or extend stronger ones.
- The summative project (see [assessments/project-brief.md](#)) runs in parallel from Day 6 onward.

Differentiation

- **For advanced students:** see [teacher-notes.md](#) for extension prompts.
- **For students needing more support:** simplify activities, do more pair-work, allow oral instead of written responses.

Homework

Homework — Magic Squares — Bhadragaṇita (Senior)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — Discover the 3×3** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — The Magic Constant** · Write 2 sentences about today's concept. (~10 min)
3. **Day 3 — How Many 3×3 Magic Squares?** · Write 2 sentences about today's concept. (~10 min)
4. **Day 4 — Stair-Step Method (Nārāyaṇa)** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — Practice — 5×5 and 7×7** · Write 2 sentences about today's concept. (~10 min)
6. **Day 6 — 4×4 and the Khajuraho Square** · Write 2 sentences about today's concept. (~10 min)
7. **Day 7 — Historical Context — Nārāyaṇa and Predecessors** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Cross-Cultural — Lo Shu and Dürer** · Write 2 sentences about today's concept. (~10 min)
9. **Day 9 — Design Your Own** · Write 2 sentences about today's concept. (~10 min)
10. **Day 10 — Final Share and Assessment** · Write 2 sentences about today's concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.